

ECO102: Introduction to Macroeconomics

Lecture 2: Measuring GDP & Economic Growth Contd.

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Recap

- ❑ GDP is the market value of the final goods and services produced within a country in a given time period
- ❑ The circular flow model is the foundational basis of the measurement of GDP
- ❑ The circular flow model outlines that: $GDP = \text{Aggregate Expenditure} = \text{Aggregate Income}$
- ❑ Hence the GDP of a nation can both be calculated using the Expenditure or the Income Approach
- ❑ $Y = C + I + G + (X - M)$
- ❑ Financial transfers are not a part of the circular flow of expenditure and income

Measuring GDP

- The Expenditure Approach vs The Income Approach
- The expenditure approach measures GDP as the sum of consumption expenditure (C), investment (I), government expenditure on goods and services (G), and net exports of goods and services ($X - M$).
- The income approach measures GDP by summing the incomes that firms pay households for the services of the factors of production they hire; wages for labor, interest for capital, rent for land, and profit for entrepreneurship.

Expenditure Approach

- ***Personal consumption expenditures (C):***
 - Expenditures by households on goods and services
 - Goods: Food items, Education items, Apparels, Household durable goods etc.
 - Services: Banking Services, Legal Services, Medical Services, etc.
 - **Question:** Will purchase of a new home be under personal consumption expenditure?
- ***Gross private domestic investment (I):***
 - Expenditure on capital equipment and buildings by firms and the additions to business inventories

Expenditure Approach

- ***Government expenditure on goods and services (G):***
 - Expenditure by all levels of government on goods and services
 - National Defense, Garbage Collection, Public Education, Public Health, Infrastructural Development
 - DOES NOT INCLUDE TRANSFER PAYMENTS
- ***Net exports of goods and services ($X - M$):***
 - The value of exports (X) minus the value of imports (M)
 - Exports: Garments industries in Bangladesh selling RMG to the US
 - Imports: A South Korean Samsung Phone being sold in Bangladesh

Examples

Item	Symbol	Amount in 2010 (Billions US\$)	Percentage of GDP
Personal Consumption Expenditure	C	10,285	70.5
Gross Private Domestic Investment	I	1,842	12.6
Government Expenditure on Goods and Services	G	2,991	20.5
Net Exports of Goods and Services	X-M	-531	-3.7
GDP	Y	14,579	100

Practice

Use the following data to calculate aggregate expenditure and imports of goods and services:

- Government expenditure: \$20 billion
- Aggregate income: \$100 billion
- Consumption expenditure: \$67 billion
- Investment: \$21 billion
- Exports of goods and services: \$30 billion

Income Approach

- ***Compensation of employees:***
 - Payment for labor service
 - Net wages and salaries (called “take-home pay”) that workers receive plus taxes withheld on earnings plus fringe benefits
- ***Net operating surplus:***
 - Sum of all other factor incomes
 - Four components: net interest, rental income, corporate profits, and proprietors’ income
- ***Indirect taxes less subsidies and Depreciation***

Nominal vs Real GDP

- ***Nominal GDP***: The value of final goods and services produced in a given year when valued at the prices of that year
- ***Real GDP***: the value of final goods and services produced in a given year when ***valued at the prices of a reference base year***
 - ***To compare the standard of living over time***
 - ***To compare the standard of living across countries***

Calculation of Real GDP

		Quantity	Price	Expenditure
Item		(millions)	(dollars)	(millions of dollars)
(a) In 2005				
C	T-shirts	10	5	50
I	Computer chips	3	10	30
G	Security services	1	20	20
Y	Real and Nominal GDP in 2005			100
(b) In 2010				
C	T-shirts	4	5	20
I	Computer chips	2	20	40
G	Security services	6	40	240
Y	Nominal GDP in 2010			300
(c) Quantities of 2010 valued at prices of 2005				
C	T-shirts	4	5	20
I	Computer chips	2	10	20
G	Security services	6	20	120
Y	Real GDP in 2010			160

Uses of Real GDP

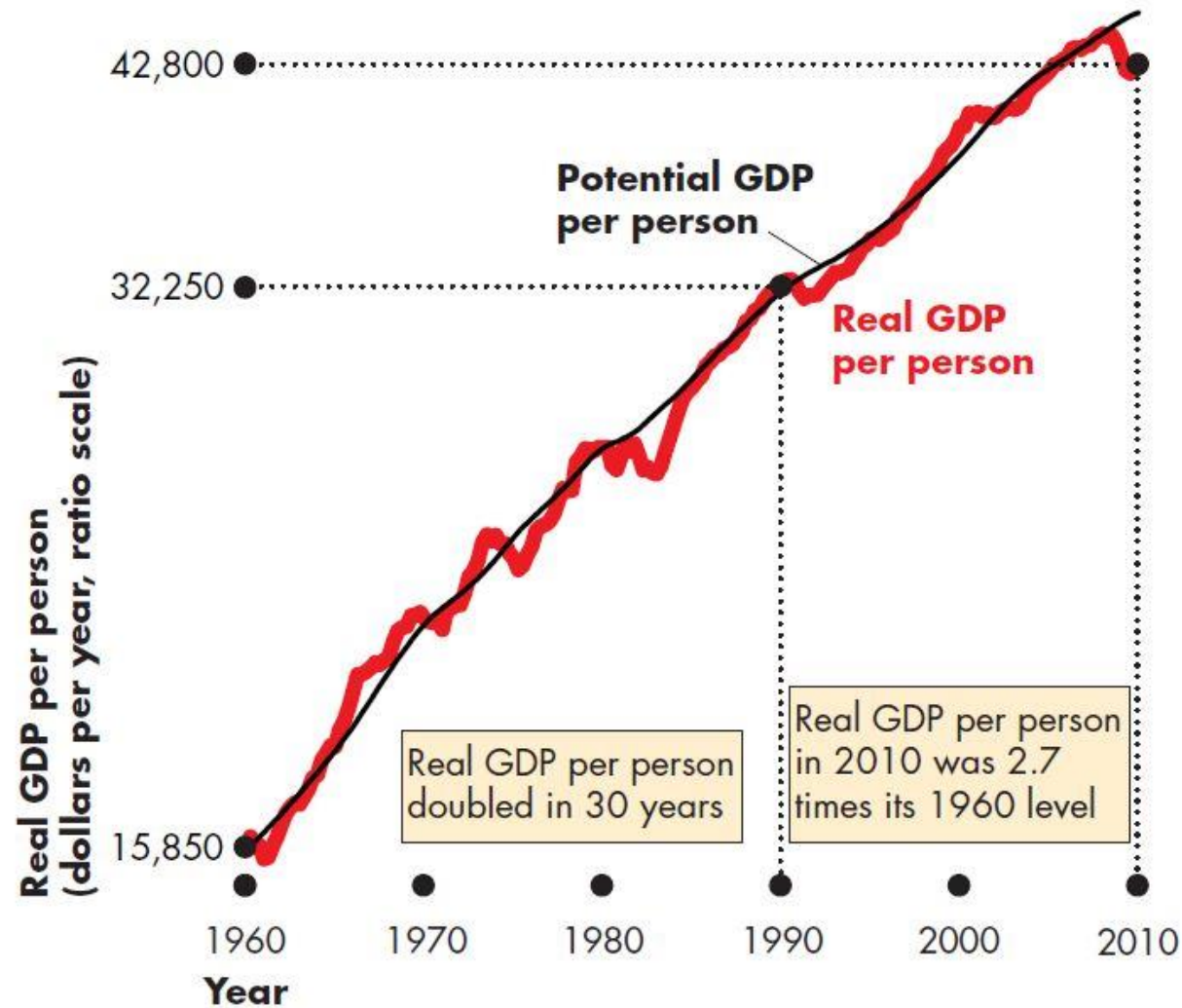
- By using real GDP, we remove any influence that rising prices and a rising cost of living might have had on our comparison
- One method of comparing the standard of living over time is to calculate real GDP per person in different years
- ***GDP Per Capita*** = Real GDP/Total Population
- We're interested in both the long-term trends and the shorter-term cycles in the standard of living

Standard of Living Over Time

- ***Long Term Trends:***

- In 1960, real GDP per person in US was \$15,850 and in 2010, it was \$42,800. So real GDP per person in 2010 was 2.7 times its 1960. To the extent that real GDP per person measures the standard of living, people were 2.7 times as well off in 2010 as their grandparents had been in 1960
- The growth of **potential GDP** per person:
 - Potential GDP is the maximum level of real GDP that can be produced while avoiding shortages of labor, capital, land, and entrepreneurial ability that would bring rising inflation
 - Potential GDP is a theoretical calculation and potential GDP per person doesn't grow at a constant pace
- Fluctuations of Real GDP

Standard of Living Over Time



Standard of Living Over Time

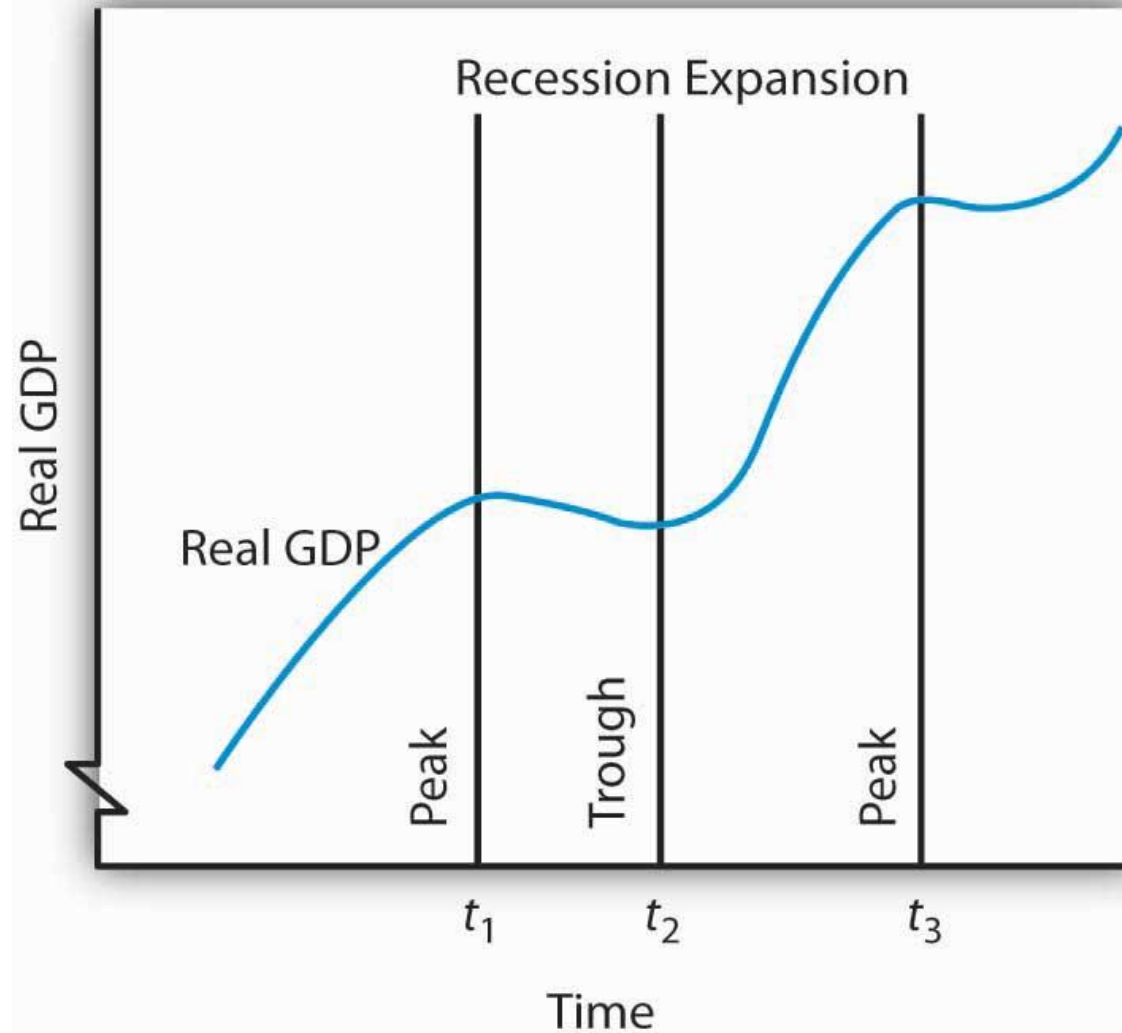
- ***Fluctuations of Real GDP:***

- The periodic fluctuation of the real GDP is primarily caused by the ***Business Cycle***

- ***The Business Cycle:***

- A periodic but irregular up-and-down movement of total production and other measures of economic activity.
- Every cycle has two phases: **1. Expansion; 2. Recession**
- And two turning points: **1. Peak; 2. Trough**

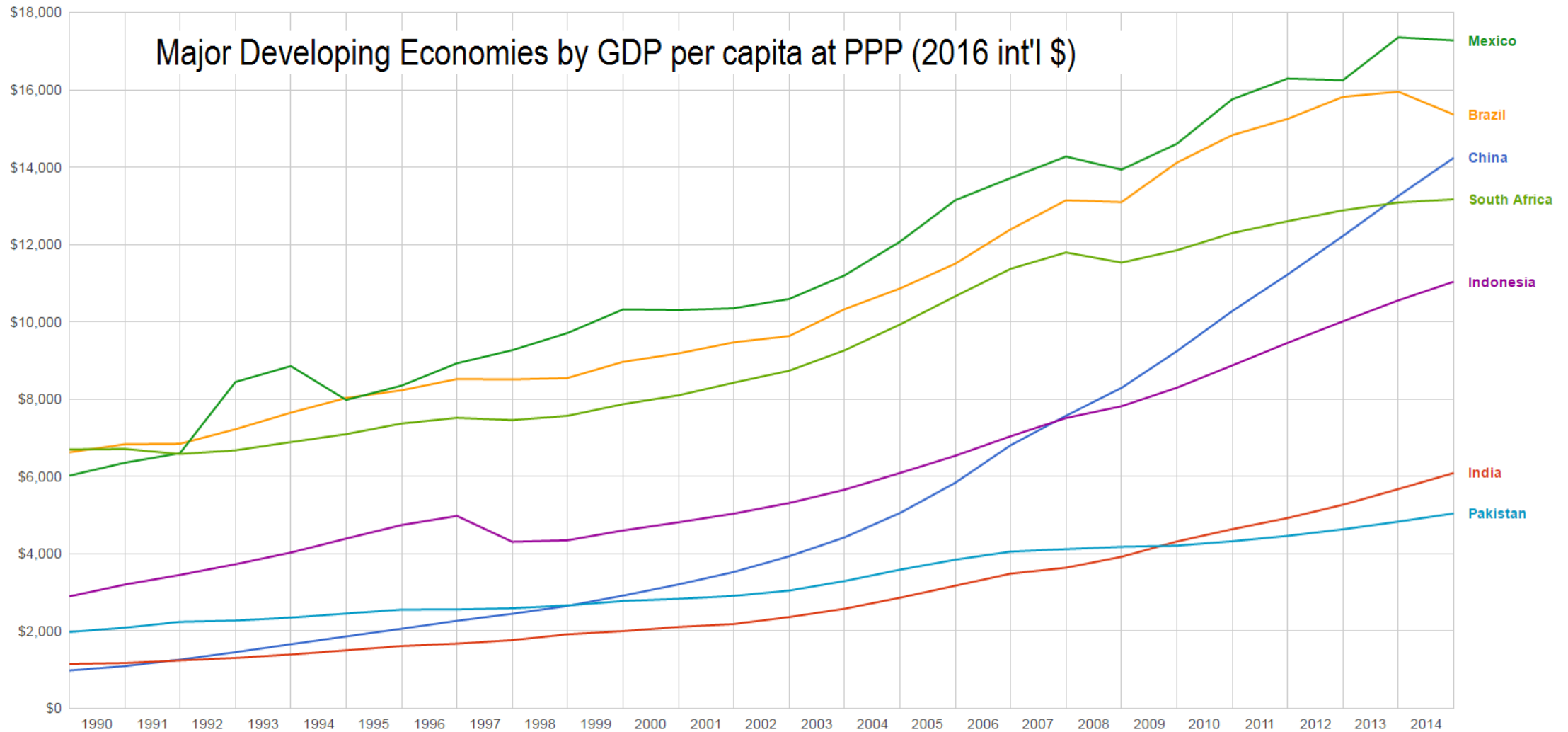
Standard of Living Over Time



Standard of Living Across Countries

- Real GDP can be used to compare the standard of living across countries
- There are two problems to this
 - The real GDP of one country must be converted into the same currency units as the real GDP of the other country
 - The goods and services in both countries must be valued at the same prices:
Using purchasing power parity or PPP

Standard of Living Across Countries



Limitations of Real GDP

- ***Household production:***

- The omission of household production from GDP means that GDP underestimates total production
- It also means that the growth rate of GDP overestimates the growth rate of total production

- ***Underground economic activity:***

- The underground economy is the part of the economy that is purposely hidden from the view of the government to avoid taxes and regulations or because the goods and services being produced are illegal

- ***Health and life expectancy:***

- Good health and a long life do not show up in real GDP

Limitations of Real GDP

- ***Leisure time:***

- Leisure time is a measurement of well-being, however it not included within the GDP

- ***Environmental quality:***

- Adverse environmental impacts from economic activities and the environmental quality in general is not reflected in the GDP

- ***Political freedom and social justice:***

- Political freedom, access to social security safety nets, equality of opportunity are not reflected in the GDP

Question?

IS GDP A GOOD MEASURE OF ECONOMIC WELL-BEING